Linear Models I

Linear Algebra · Lecture 11

Last time: $z \sim N_n(\mu, \Sigma) \Rightarrow Az + b \sim N_m(A\mu + b, A\Sigma A')$ (Lecture 10, Theorem 3); $z \sim N_n(\mu, \sigma^2 I_n)$, $A' = A = A^2$, $\operatorname{rank} A = r \geqslant 1$, $A\mu = 0 \Rightarrow z'Az/\sigma^2 \sim \chi^2_r$ (Lecture 10, Theorem 15). Earlier results are cited where they are used. e is the vector of residuals; unit vectors are written as columns of I.

1. The linear model and least squares

Definition 1. In the **linear model** $y = X\beta + \varepsilon$, the **design matrix** $X \in \mathbb{R}^{n\times p}$ is a known constant matrix, $\beta \in \mathbb{R}^p$ is an unknown constant vector of **parameters**, the **errors** $\varepsilon \in \mathbb{R}^n$ form a random vector, and the **response** $y \in \mathbb{R}^n$ is observed; x_i' is row i of X. In **multiple regression**, $p = k + 1$, $\beta = (\beta_0, \ldots, \beta_k)'$ and $x_i' = (1, x_{i1}, \ldots, x_{ik})$, with x_{ij} the value of the jth explanatory variable on observation i: the first column $\mathbf{1}_n = (1, \ldots, 1)'$ of X (Lecture 1, Definition 14) is the column of the **intercept** β_0. The assumptions: (A1) $\mathsf{E}(y) = X\beta$; (A2) $\mathsf{E}(\varepsilon) = 0$; (A3) $\operatorname{var}(\varepsilon) = \sigma^2 I_n$ ($\sigma^2 > 0$); (A4) $\operatorname{rank} X = p$. Results hold for X of any rank unless (A4) is assumed. A **least-squares estimate** of β is a $\hat\beta \in \mathbb{R}^p$ that minimizes $S(\beta) = \|y - X\beta\|^2 = \sum_i (y_i - x_i'\beta)^2$; the p equations $X'X\beta = X'y$ are the **normal equations**.

Here ε, y are random vectors (Lecture 9, Definition 1: vectors of random variables). By Lecture 9, Theorem 3(b) ($\mathsf{E}(Az+b) = A\,\mathsf{E}(z)+b$, $\operatorname{var}(Az+b) = A\operatorname{var}(z)A'$; $A = I_n$, $b = X\beta$): $\mathsf{E}(y) = X\beta + \mathsf{E}(\varepsilon)$, $\operatorname{var}(y) = \operatorname{var}(\varepsilon)$. So (A1) $\iff$ (A2), and (A3) gives $\operatorname{var}(y) = \sigma^2 I_n$. (A4) needs $n \geqslant p$ (Lecture 4, Theorem 12(a): $\operatorname{rank} X \leqslant \min\{n, p\}$). An **estimator** $T = g(y)$ of a parameter θ is a random variable, g involving no unknown parameter; it is **unbiased** if $\mathsf{E}(T) = \theta$ for all values of the parameters (Lecture 9, Definition 7).

Theorem 2. Let $X \in \mathbb{R}^{n\times p}$ and $y \in \mathbb{R}^n$.

(a) The normal equations $X'X\beta = X'y$ have a solution.

(b) $X'X$ is invertible $\iff$ $\operatorname{rank} X = p$. Then $\hat\beta = (X'X)^{-1}X'y$ is the only solution.

(c) If $\hat\beta$ is a solution, the solutions are the vectors $\hat\beta + z$, $z \in \mathcal{N}(X)$; all give the same $X\hat\beta$. If $\operatorname{rank} X < p$, there are infinitely many solutions.

Proof.

(a) $\mathcal{C}(A) = \{Ax\}$, $\mathcal{N}(A) = \{x : Ax = 0\}$ (Lecture 4, Definition 7):

$$X'y \in \mathcal{C}(X') = \mathcal{C}(X'X) \qquad \text{(Lecture 4, Theorem 16(c))}$$
$$\Longrightarrow X'X\beta = X'y \text{ has a solution} \qquad \text{(Lecture 4, Corollary 13(a))}$$

(b) $X'X \in \mathbb{R}^{p\times p}$:

$$X'X \text{ invertible} \iff \operatorname{rank}(X'X) = p \iff \operatorname{rank} X = p \qquad \text{(Lecture 4, Theorems 12(d), 16(b))}$$
$$X'X\beta = X'y \implies \beta = (X'X)^{-1}X'X\beta = (X'X)^{-1}X'y, \text{ and this } \beta \text{ is a solution}$$

(c) If $Ax_p = b$, the solutions of $Ax = b$ are the $x_p + z$, $Az = 0$ (Lecture 2, Theorem 22(a)):

$$\text{solutions } \hat\beta + z,\ z \in \mathcal{N}(X'X) = \mathcal{N}(X) \qquad \text{(Lecture 4, Theorem 16(a))}$$
$$X(\hat\beta + z) = X\hat\beta + Xz = X\hat\beta$$
$$\operatorname{rank} X < p \implies \dim \mathcal{N}(X) = p - \operatorname{rank} X \geqslant 1 \qquad \text{(Lecture 4, Theorem 11)}$$
$$\implies \text{some } z \neq 0 \text{ in } \mathcal{N}(X) \qquad \text{(Lecture 4, Theorem 4: } \dim\{0\} = 0)$$
$$\hat\beta + t_1 z = \hat\beta + t_2 z \implies t_1 = t_2 : \text{ the solutions } \hat\beta + tz\ (t \in \mathbb{R}) \text{ are distinct}$$

□

Theorem 3. Let $X \in \mathbb{R}^{n\times p}$ and $y \in \mathbb{R}^n$.

(a) $\hat\beta$ is a least-squares estimate $\iff X'X\hat\beta = X'y$. For every solution $\hat\beta$ of the normal equations: $S(\beta) = S(\hat\beta) + \|X(\hat\beta - \beta)\|^2$ $(\beta \in \mathbb{R}^p)$.

(b) All least-squares estimates give the same vector of **fitted values** $\hat y = X\hat\beta$, the orthogonal projection of y onto $\mathcal{C}(X)$; the vector of **residuals** $e = y - \hat y$ lies in $\mathcal{C}(X)^\perp$.

(c) If $\operatorname{rank} X = p$: $\hat\beta = (X'X)^{-1}X'y$ is the only least-squares estimate, $\hat y = Hy$ and $e = My$, with the **hat matrix** $H = X(X'X)^{-1}X'$ (Lecture 8, Theorem 3: $P = A(A'A)^{-1}A'$, here $A = X$) and the **residual maker** $M = I_n - H$.

Proof. $y \perp \mathcal{C}(A) \iff A'y = 0$ (Lecture 6, Theorem 5(a)); the orthogonal projection of y onto W: the $\hat w \in W$ with $y - \hat w \perp W$ (Lecture 8, Definition 1); $\|x\| = 0 \iff x = 0$ (Lecture 6, Definition 1).

(a) $\Leftarrow$: $X'X\hat\beta = X'y$; $X\beta \in \mathcal{C}(X)$ for $\beta \in \mathbb{R}^p$. $\Rightarrow$: β^* a least-squares estimate, $\hat\beta$ a solution (Theorem 2(a)):

$$X'(y - X\hat\beta) = 0 \implies y - X\hat\beta \perp \mathcal{C}(X) \qquad \text{(Lecture 6, Theorem 5(a))}$$
$$\implies X\hat\beta \text{ is the projection of } y \text{ onto } \mathcal{C}(X) \qquad \text{(Lecture 8, Definition 1)}$$
$$\|y - X\beta\|^2 = \|y - X\hat\beta\|^2 + \|X\hat\beta - X\beta\|^2 \implies S(\beta) \geqslant S(\hat\beta) \qquad \text{(Lecture 8, Theorem 2)}$$
$$\|X(\hat\beta - \beta^*)\|^2 = S(\beta^*) - S(\hat\beta) \leqslant 0 \implies X\beta^* = X\hat\beta \qquad (\beta^* \text{ minimizes } S\text{; Lecture 6, Definition 1})$$
$$X'X\beta^* = X'X\hat\beta = X'y$$

(b) By the last two lines of (a), every least-squares estimate β^* gives $X\beta^* = X\hat\beta$, the projection of the first two lines; and $e = y - X\hat\beta \perp \mathcal{C}(X)$, so $e \in \mathcal{C}(X)^\perp$ (Lecture 6, Definition 4: $W^\perp = \{y : y \perp W\}$).

(c) By Theorem 2(b) and (a), $(X'X)^{-1}X'y$ is the only least-squares estimate; then $\hat y = X(X'X)^{-1}X'y = Hy$ and $e = y - Hy = My$. □

Proposition 4. Let $\hat\beta$ be a least-squares estimate, $\hat y = X\hat\beta$, $e = y - \hat y$. (a) $X'e = 0$. (b) If $\mathbf{1}_n \in \mathcal{C}(X)$: $\mathbf{1}'e = 0$, so $\hat y$ and y have the same mean $\bar y$ (the mean of $x \in \mathbb{R}^n$ is $\bar x = \frac{1}{n}\mathbf{1}'x$, Lecture 1, Example 16). (c) $\hat y'e = 0$.

Proof. (a) $X'(y - X\hat\beta) = 0$ is the normal equations (Theorem 3(a)).

(b) $\mathbf{1} = Xc$: $\mathbf{1}'e = c'X'e = 0$ by (a), and $\mathbf{1}'\hat y = \mathbf{1}'y - \mathbf{1}'e = \mathbf{1}'y$.

(c) $\hat y'e = \hat\beta'X'e = 0$ by (a). □

Example 5. Let $X = [\mathbf{1}_4\ (2,0,2,0)'\ (3,3,1,1)'] \in \mathbb{R}^{4\times 3}$ $(k = 2)$ and $y = (9,3,5,3)'$: the A and y of Lecture 8, Example 6, where (with $H = P$)

$$X'X = \begin{pmatrix} 4 & 4 & 8 \\ 4 & 8 & 8 \\ 8 & 8 & 20 \end{pmatrix}, \quad (X'X)^{-1} = \frac{1}{4}\begin{pmatrix} 6 & -1 & -2 \\ -1 & 1 & 0 \\ -2 & 0 & 1 \end{pmatrix}, \quad (X'X)^{-1}X' = \frac{1}{4}\begin{pmatrix} -2 & 0 & 2 & 4 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \end{pmatrix}$$

and $Hy = (8,4,6,2)'$, $(I - H)y = (1,-1,-1,1)'$. Find $\hat\beta$, $\hat y$, e, $S(\hat\beta)$ and $S((1,2,2)')$.

Solution.

$X'X$ is invertible, so $\operatorname{rank} X = 3 = p$ (Theorem 2(b)): (A4) holds; $X'y$ has the entries $\operatorname{col}_j(X)'y$ (Lecture 1, Theorem 4(d), Definition 7):

$$X'y = \begin{pmatrix} 9+3+5+3 \\ 18+10 \\ 27+9+5+3 \end{pmatrix} = \begin{pmatrix} 20 \\ 28 \\ 44 \end{pmatrix}, \qquad \hat\beta = (X'X)^{-1}X'y = \frac{1}{4}\begin{pmatrix} -18+0+10+12 \\ 9-3+5-3 \\ 9+3-5-3 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$$

$$X'X\hat\beta = (4+8+8,\ 4+16+8,\ 8+16+20)' = (20,28,44)' = X'y$$

Theorem 3; Proposition 4 for the checks; the identity of Theorem 3(a) for $S((1,2,2)')$:

$$\hat{y} = X\hat{\beta} = (1+4+3,\ 1+0+3,\ 1+4+1,\ 1+0+1)' = (8,4,6,2)' = Hy$$
$$e = y - \hat{y} = (1,-1,-1,1)'$$
$$S(\hat{\beta}) = e'e = 1+1+1+1 = 4 > 0 :\ X\beta = y \text{ has no solution}$$
$$X'e = (1-1-1+1,\ 2-2,\ 3-3-1+1)' = 0, \qquad \mathbf{1}'e = 0, \qquad \hat{y}'e = 8-4-6+2 = 0$$
$$S((1,2,2)') = 4 + \|X(0,0,-1)'\|^2 = 4 + (9+9+1+1) = 24$$

2. Rank-deficient designs and generalized inverses

Theorem 6. Let $n_1, \dots, n_g \geqslant 1$, $n = n_1 + \cdots + n_g$; n observations in g groups (first the n_1 of group 1, then the n_2 of group 2, and so on); the **indicator** $d_l \in \mathbb{R}^n$ of group l has entry 1 in the rows of group l and 0 elsewhere. The model $y_i = \mu + \alpha_l + \varepsilon_i$ (observation i in group l) is $y = X\beta + \varepsilon$, $\beta = (\mu, \alpha_1, \dots, \alpha_g)'$, $X = [\mathbf{1}_n\ d_1\ \cdots\ d_g] \in \mathbb{R}^{n\times(g+1)}$. Let $X_1 = [d_1\ \cdots\ d_g]$, $X_2 = [\mathbf{1}_n\ d_2\ \cdots\ d_g] \in \mathbb{R}^{n\times g}$.

(a) X_1 has full column rank, and $X = X_1 S$ with $S = [\mathbf{1}_g\ I_g] \in \mathbb{R}^{g\times(g+1)}$ of full row rank.

(b) $\operatorname{rank} X = g$ and $\mathcal{N}(X) = \operatorname{span}\{(-1,1,\dots,1)'\}$.

(c) $X_1 = XT_1$, $X_2 = XT_2$, with $T_1, T_2 \in \mathbb{R}^{(g+1)\times g}$ the matrix I_{g+1} without its first, resp. second, column. X_1, X_2 have full column rank g, and $\mathcal{C}(X_1) = \mathcal{C}(X_2) = \mathcal{C}(X)$.

Proof. $A \in \mathbb{R}^{m\times n}$ has full column rank $\iff$ its columns are independent $\iff \mathcal{N}(A) = \{0\}$, and full row rank $\iff \mathcal{C}(A) = \mathbb{R}^m$ (Lecture 4, Theorem 12(b),(c)).

(a) No observation lies in two groups, each lies in one:

$$d_l'd_m = 0\ (l \neq m), \qquad d_l'd_l = n_l$$
$$c_1d_1 + \cdots + c_gd_g = 0 \implies c_m n_m = 0 \text{ (multiply by } d_m') \implies c_m = 0$$
$$\implies X_1 \text{ has full column rank} \qquad \text{(Lecture 4, Theorem 12(b))}$$
$$d_1 + \cdots + d_g = \mathbf{1}_n \implies X_1\mathbf{1}_g = \mathbf{1}_n \qquad \text{(Lecture 1, Theorem 4(b))}$$
$$\implies X_1 S = [X_1\mathbf{1}_g\ \ X_1 I_g] = X \qquad \text{(Lecture 1, Theorem 4(a))}$$
$$\mathcal{C}(S) \supseteq \mathcal{C}(I_g) = \mathbb{R}^g :\ S \text{ full row rank} \qquad \text{(Lecture 4, Theorem 12(c))}$$

(b) X_1 has full column rank ((a)), so $\operatorname{rank}(X_1B) = \operatorname{rank} B$, $\mathcal{N}(X_1B) = \mathcal{N}(B)$ (Lecture 4, Theorem 15(b)):

$$\operatorname{rank} X = \operatorname{rank}(X_1S) = \operatorname{rank} S = g \qquad \text{(Lecture 4, Theorem 15(b); (a))}$$
$$\mathcal{N}(X) = \mathcal{N}(X_1S) = \mathcal{N}(S) \qquad \text{(Lecture 4, Theorem 15(b))}$$
$$S(c_0, c_1, \dots, c_g)' = c_0\mathbf{1}_g + (c_1, \dots, c_g)' = 0 \iff c_l = -c_0\ (l = 1, \dots, g)$$
$$\iff (c_0, c_1, \dots, c_g)' = -c_0(-1, 1, \dots, 1)'$$

(c)

- $X\operatorname{col}_j(T_1)$ is a column of X other than the first (Lecture 1, Theorem 4(a),(b)), so $XT_1 = X_1$; likewise $XT_2 = X_2$.
- $\mathbf{1}_n = d_1 + \cdots + d_g$ and $d_1 = \mathbf{1}_n - d_2 - \cdots - d_g$: each column of X_2 lies in $\mathcal{C}(X_1)$ and each column of X_1 in $\mathcal{C}(X_2)$; so $\mathcal{C}(X_2) = \mathcal{C}(X_1)$, of dimension g by (a), and X_2, with g columns, has full column rank (Lecture 4, Definition 10: $\operatorname{rank} A = \dim \mathcal{C}(A)$).
- $\mathcal{C}(X_1) = \mathcal{C}(XT_1) \subseteq \mathcal{C}(X)$ (Lecture 4, Theorem 15(a)), both of dimension g ((b)): equal (Lecture 4, Theorem 5(a): a subspace $U \subseteq W$ with $\dim U = \dim W$ is W).

□

So the intercept together with all g indicators (the **dummy variable trap**) makes $X'X$ singular (Theorem 2(b)); dropping the intercept or one indicator (d_1 in Theorem 6(c); any d_l after renumbering

the groups) gives a design of full column rank with the same column space, hence the same fitted values (Theorem 3(b)).

Example 7. Six plants of three types weigh $5, 7, 6$ (type 1), $3, 5$ (type 2) and 2 (type 3): $y = (5, 7, 6, 3, 5, 2)'$, the model of Theorem 6 with $g = 3$, $(n_1, n_2, n_3) = (3, 2, 1)$, and (entries $\mathrm{col}_j(X)'\,\mathrm{col}_l(X)$, $\mathrm{col}_j(X)'y$: Lecture 1, Theorem 4(d), Definition 7)

$$X = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}, \qquad X'X = \begin{pmatrix} 6 & 3 & 2 & 1 \\ 3 & 3 & 0 & 0 \\ 2 & 0 & 2 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}, \qquad X'y = \begin{pmatrix} 28 \\ 18 \\ 8 \\ 2 \end{pmatrix}$$

Find all least-squares estimates and $\hat{y}$, and the estimates with X_1 and with X_2.

Solution.

$\operatorname{rank} X = 3$, $\mathcal{N}(X) = \operatorname{span}\{(-1, 1, 1, 1)'\}$ (Theorem 6(b)): $X'X$ is singular (Theorem 2(b)); all solutions, that is, all least-squares estimates, by Theorems 2(c) and 3(a):

$$\begin{aligned} X'X(0, 6, 4, 2)' &= (18 + 8 + 2,\ 18,\ 8,\ 2)' = (28, 18, 8, 2)' = X'y \\ \hat{\beta} &= (0, 6, 4, 2)' + t(-1, 1, 1, 1)' = (-t,\ 6 + t,\ 4 + t,\ 2 + t)' \quad (t \in \mathbb{R}) \\ X\hat{\beta} &= (6, 6, 6, 4, 4, 2)' \text{ for all } t: \text{ the group means } 18/3,\ 8/2,\ 2/1 \\ \hat{\alpha}_1 - \hat{\alpha}_2 &= (6 + t) - (4 + t) = 2 \text{ for all } t; \qquad \hat{\alpha}_1 = 6 + t \text{ changes with } t \end{aligned}$$

X_1, X_2 have full column rank and $\mathcal{C}(X_1) = \mathcal{C}(X_2) = \mathcal{C}(X)$ (Theorem 6(c)): one solution each (Theorem 2(b)), the same $\hat{y}$ (Theorem 3(b)):

$$X_1'X_1 = \operatorname{diag}(3, 2, 1), \qquad X_1'y = (18, 8, 2)', \qquad \hat{\beta} = (X_1'X_1)^{-1}X_1'y = (6, 4, 2)'$$

$$X_2'X_2 = \begin{pmatrix} 6 & 2 & 1 \\ 2 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \qquad X_2'y = \begin{pmatrix} 28 \\ 8 \\ 2 \end{pmatrix}, \qquad X_2'X_2 \begin{pmatrix} 6 \\ -2 \\ -4 \end{pmatrix} = \begin{pmatrix} 36 - 4 - 4 \\ 12 - 4 \\ 6 - 4 \end{pmatrix} = X_2'y$$

$$\hat{y} = 6d_1 + 4d_2 + 2d_3 = 6\,\mathbf{1}_6 - 2d_2 - 4d_3 = (6, 6, 6, 4, 4, 2)'$$

$(6, 4, 2)'$ and $(6, -2, -4)'$ are the coordinates of $\hat{y}$ in the bases d_1, d_2, d_3 and $\mathbf{1}_6, d_2, d_3$ of $\mathcal{C}(X)$ (Lecture 3, Theorem 15: the coordinates in a basis are unique). In this **baseline coding**, the coefficient of $\mathbf{1}_6$ is the value of group 1, and those of d_2, d_3 are the differences of groups 2, 3 from it. $T_1(6, 4, 2)' = (0, 6, 4, 2)'$ and $T_2(6, -2, -4)' = (6, 0, -2, -4)'$ are the solutions with $t = 0$ and $t = -6$.

Definition 8. Let $A \in \mathbb{R}^{m \times n}$. A matrix $G \in \mathbb{R}^{n \times m}$ with $AGA = A$ is a **generalized inverse** (**g-inverse**) of A; A^- denotes any g-inverse of A. If $y = Az \in \mathcal{C}(A)$, then $A(Gy) = AGAz = Az = y$: $x = Gy$ solves $Ax = y$. If $A \in \mathbb{R}^{n \times n}$ is invertible, A^{-1} is its only g-inverse: $AGA = A$ gives $G = A^{-1}(AGA)A^{-1} = A^{-1}AA^{-1} = A^{-1}$.

Theorem 9. Let $A \in \mathbb{R}^{m\times n}$, $\operatorname{rank} A = r \geqslant 1$. For rows $i_1 < \cdots < i_r$ and columns $j_1 < \cdots < j_r$, let the submatrix $A_* \in \mathbb{R}^{r\times r}$ ((s,t) entry $a_{i_s j_t}$) be invertible, and let $G \in \mathbb{R}^{n\times m}$ have (j_s, i_t) entry equal to the (s,t) entry of A_*^{-1} $(s,t = 1,\ldots,r)$ and all other entries 0. Then G is a g-inverse of A. In particular, if $A = \left(\begin{smallmatrix} A_{11} & A_{12} \\ A_{21} & A_{22}\end{smallmatrix}\right)$ with $A_{11} \in \mathbb{R}^{r\times r}$ invertible, then $\left(\begin{smallmatrix} A_{11}^{-1} & 0 \\ 0 & 0\end{smallmatrix}\right) \in \mathbb{R}^{n\times m}$ is a g-inverse of A.

Proof.

Selection matrices $U = [\operatorname{col}_{i_1}(I_m) \ \cdots \ \operatorname{col}_{i_r}(I_m)]$ and $V = [\operatorname{col}_{j_1}(I_n) \ \cdots \ \operatorname{col}_{j_r}(I_n)]$; the (s,t) entry of $U'AV$ is $\operatorname{col}_{i_s}(I_m)'A\operatorname{col}_{j_t}(I_n) = a_{i_s j_t}$ (Lecture 1, Theorem 4(d), Proposition 11(a)):

$$
\begin{aligned}
U'AV = A_*, \qquad B = AV \in \mathbb{R}^{m\times r}, \qquad & \mathcal{C}(B) \subseteq \mathcal{C}(A) && \text{(Lecture 4, Theorem 15(a))}\\
Bz = 0 \implies A_* z = U'Bz = 0 \implies & z = 0 &&\\
\implies B \text{ has independent columns} & && \text{(Lecture 4, Theorem 12(b))}\\
\dim \mathcal{C}(A) = r \implies \text{these } r \text{ columns form a basis of } & \mathcal{C}(A) && \text{(Lecture 4, Theorem 5(d))}\\
\implies A = BT,\ T \in \mathbb{R}^{r\times n} & && \text{(Lecture 1, Theorem 4(a),(b))}\\
U'A = U'BT = A_*T \implies & T = A_*^{-1}U'A &&\\
A(VA_*^{-1}U')A = BA_*^{-1}U'A = BT = & A &&\\
VA_*^{-1}U' = \textstyle\sum_{s,t}(A_*^{-1})_{st}\operatorname{col}_{j_s}(I_n)\operatorname{col}_{i_t}(I_m)' = & G && \text{(Lecture 1, Theorem 4(d))}
\end{aligned}
$$

□

The case $A_* = A_{11}$ is $i_s = j_s = s$ $(s = 1,\ldots,r)$. So: pick an invertible $r \times r$ submatrix, invert it, put A_*^{-1} (not transposed) into rows $j_1,\ldots,j_r$ and columns $i_1,\ldots,i_r$ of the $n \times m$ zero matrix.

Theorem 10. Let $X \in \mathbb{R}^{n\times p}$, $r = \operatorname{rank} X$, G a g-inverse of $X'X$, and $H = XGX' = X(X'X)^-X'$, $M = I_n - H$ (the **hat matrix** and the **residual maker**).

(a) $XGX'X = X$ and $X'XGX' = X'$; $\beta^0 = GX'y$ is a solution of the normal equations.

(b) For every $y \in \mathbb{R}^n$, Hy is the orthogonal projection of y onto $\mathcal{C}(X)$. So H does not depend on the choice of G, and $H = X(X'X)^{-1}X'$ if $r = p$.

(c) $H' = H = H^2$, $HX = X$, $X'H = X'$; $M' = M = M^2$, $MX = 0$, $HM = MH = 0$.

(d) $\operatorname{rank} H = \operatorname{tr} H = r$ and $\operatorname{rank} M = \operatorname{tr} M = n - r$.

(e) For every least-squares estimate $\hat\beta$: $\hat y = Hy$, $e = My$, and the **error sum of squares** is $\mathrm{SSE} = S(\hat\beta) = e'e = y'My = y'y - \hat\beta' X'y$.

Proof.

(a) $X'X(GX'X - I_p) = X'XGX'X - X'X = 0$ (Definition 8), so the columns of $GX'X - I_p$ lie in $\mathcal{N}(X'X)$ (Lecture 1, Theorem 4(a)):

$$
\begin{aligned}
\mathcal{N}(X'X) = \mathcal{N}(X) \implies X(GX'X - I_p) = 0:\ XGX'X = X && \text{(Lecture 4, Theorem 16(a))}\\
X'XG'X'X = (X'XGX'X)' = (X'X)' = X'X && \text{(Lecture 1, Theorem 8)}\\
\implies G' \text{ is a g-inverse of } X'X &&\\
XG'X'X = X \text{ (line 1 for } G') \implies X'XGX' = (XG'X'X)' = X' && \text{(Lecture 1, Theorem 8)}
\end{aligned}
$$

$X'y \in \mathcal{C}(X'X)$ (Theorem 2(a)), so $X'X(GX'y) = X'y$ (Definition 8).

(b) By (a), Lecture 6, Theorem 5(a) and Lecture 8, Definition 1 (the projection is unique):

$$
\begin{aligned}
Hy = X(GX'y) \in \mathcal{C}(X), \qquad & X'(y - Hy) = X'y - X'XGX'y = 0\\
\implies Hy \text{ is the projection of } y \text{ onto } & \mathcal{C}(X)\\
G_1, G_2 \text{ g-inverses} \implies XG_1X'y = XG_2X'y \text{ (all } & y)\\
\implies XG_1X' = XG_2X' \qquad & \text{(Lecture 1, Proposition 11(b))}
\end{aligned}
$$

If $r = p$: $X'X$ is invertible (Theorem 2(b)), $(X'X)^{-1}$ is its only g-inverse (Definition 8), and $H = X(X'X)^{-1}X'$.

(c) G' is a g-inverse of $X'X$ ((a)), so $H = XG'X'$ ((b)):

$$H' = (XGX')' = XG'X' = H \qquad \text{(Lecture 1, Theorem 8)}$$
$$HX = XGX'X = X, \qquad X'H = X'H' = (HX)' = X' \qquad \text{((a); Lecture 1, Theorem 8)}$$
$$H^2 = (HX)GX' = XGX' = H$$
$$M' = I - H' = M, \qquad M^2 = I - 2H + H^2 = M$$
$$MX = X - HX = 0, \qquad HM = H - H^2 = 0 = MH$$

(d) H, M are symmetric and idempotent ((c)):

$$\mathcal{C}(H) = \mathcal{C}(X(GX')) \subseteq \mathcal{C}(X) = \mathcal{C}(HX) \subseteq \mathcal{C}(H) \qquad \text{(Lecture 4, Theorem 15(a))}$$
$$\operatorname{rank} H = \dim \mathcal{C}(X) = r \qquad \text{(Lecture 4, Definition 10)}$$
$$\operatorname{rank} H = \operatorname{tr} H, \qquad \operatorname{rank} M = \operatorname{tr} M \qquad \text{(Lecture 8, Theorem 11)}$$
$$\operatorname{tr} M = \operatorname{tr} I_n - \operatorname{tr} H = n - r \qquad \text{(Lecture 1, Theorem 10(a))}$$

(e) $\hat{y} = X\hat{\beta}$ is the projection of y onto $\mathcal{C}(X)$ (Theorem 3(b)), which is Hy by (b); so $e = y - Hy = My$. By (c), $S(\hat{\beta}) = e'e = y'M'My = y'My$; and $y'Hy = y'X\hat{\beta} = \hat{\beta}'X'y$ (Lecture 1, Theorem 8), so $y'My = y'y - \hat{\beta}'X'y$. □

Example 11. In Example 7, find a g-inverse G of $X'X$ by Theorem 9, $\beta^0 = GX'y$ and SSE.

Solution.

$\operatorname{rank}(X'X) = \operatorname{rank} X = 3$ (Lecture 4, Theorem 16(b)); rows and columns $2, 3, 4$ of $X'X$ give $A_* = \operatorname{diag}(3, 2, 1)$, $A_*^{-1} = \operatorname{diag}(\frac{1}{3}, \frac{1}{2}, 1)$ (Lecture 1, Proposition 13(b); Lecture 2, Theorem 15):

$$G = \operatorname{diag}(0, \tfrac{1}{3}, \tfrac{1}{2}, 1) \qquad \text{(Theorem 9)}$$
$$\beta^0 = GX'y = (0,\ 18/3,\ 8/2,\ 2)' = (0, 6, 4, 2)' \qquad \text{(Theorem 10(a))}$$
$$\text{SSE} = y'y - \beta^{0\prime}X'y = 148 - (0 + 108 + 32 + 4) = 4 \qquad \text{(Theorem 10(e))}$$
$$e = y - X\beta^0 = (-1, 1, 0, -1, 1, 0)', \qquad e'e = 4$$

Here $y'y = 25 + 49 + 36 + 9 + 25 + 4 = 148$, and β^0 is the solution with $t = 0$ in Example 7.

3. The Gauss–Markov theorem

Theorem 12. Assume (A4), and let $\hat{\beta} = (X'X)^{-1}X'y$.

(a) $\hat{\beta} = \beta + (X'X)^{-1}X'\varepsilon$; under (A2), $\mathsf{E}(\hat{\beta}) = \beta$.

(b) Under (A3), $\operatorname{var}(\hat{\beta}) = \sigma^2(X'X)^{-1}$.

Proof.

(a) $y = X\beta + \varepsilon$:

$$\hat{\beta} = (X'X)^{-1}X'(X\beta + \varepsilon) = \beta + (X'X)^{-1}X'\varepsilon \qquad \text{(Lecture 1, Theorem 3(a))}$$
$$\mathsf{E}(\hat{\beta}) = \beta + (X'X)^{-1}X'\,\mathsf{E}(\varepsilon) = \beta \qquad \text{(Lecture 9, Theorem 3(b), } z = \varepsilon, b = \beta\text{; (A2))}$$

(b) $((X'X)^{-1})' = (X'X)^{-1}$ (Lecture 1, Theorem 8; Lecture 2, Corollary 12); by (a):

$$\operatorname{var}(\hat{\beta}) = (X'X)^{-1}X'(\sigma^2 I_n)X(X'X)^{-1} = \sigma^2(X'X)^{-1} \qquad \text{(Lecture 9, Theorem 3(b); (A3))}$$

□

Theorem 13 (Gauss–Markov)**.** Assume (A1)–(A4), and let $\hat\beta = (X'X)^{-1}X'y$. Let $L \in \mathbb{R}^{p\times n}$ be constant, and $\tilde\beta = Ly$ a **linear unbiased estimator** of β: $\mathsf{E}(Ly) = \beta$ for every $\beta \in \mathbb{R}^p$. Then $\operatorname{var}(\tilde\beta) - \operatorname{var}(\hat\beta)$ is PSD (Lecture 8, Definition 13: symmetric, $x'Ax \geqslant 0$ for all x), and it is 0 only if $\tilde\beta = \hat\beta$. So $\operatorname{var}(a'\tilde\beta) \geqslant \operatorname{var}(a'\hat\beta)$ for every $a \in \mathbb{R}^p$; in particular $\operatorname{var}(\tilde\beta_j) \geqslant \operatorname{var}(\hat\beta_j)$ for each entry. $\hat\beta$ is the **BLUE** (best linear unbiased estimator) of β.

Proof.

$LX = I_p$ (Lecture 9, Theorem 3(b), (A1)); the difference D:

$$\begin{aligned}
\mathsf{E}(Ly) = L\,\mathsf{E}(y) = LX\beta = \beta \text{ for all } \beta & \\
\beta = \operatorname{col}_j(I_p)\ (j = 1,\dots,p) \implies LX = I_p & \quad \text{(Lecture 1, Theorem 4(b))}\\
D = L - (X'X)^{-1}X' \implies DX = LX - I_p = 0 & \quad \text{(Lecture 1, Theorem 3(a))}\\
X'D' = (DX)' = 0 & \quad \text{(Lecture 1, Theorem 8)}
\end{aligned}$$

The variances (Lecture 9, Theorem 3(b), (A3)); $DX = 0$, $X'D' = 0$, $(X'X)^{-1}$ symmetric (Lecture 2, Corollary 12); $(D')'D'$ is PSD (Lecture 8, Theorem 15(b)):

$$\begin{aligned}
\operatorname{var}(\tilde\beta) = L(\sigma^2 I_n)L' = \sigma^2 LL' & \\
LL' = (X'X)^{-1}X'X(X'X)^{-1} + (X'X)^{-1}X'D' & \\
\quad + DX(X'X)^{-1} + DD' = (X'X)^{-1} + DD' & \quad \text{(Lecture 1, Theorems 3(a), 8)}\\
\operatorname{var}(\tilde\beta) - \operatorname{var}(\hat\beta) = \sigma^2 DD' = \sigma^2 (D')'D' \text{ is PSD} & \quad \text{(Theorem 12(b))}\\
\sigma^2 DD' = 0 \implies \operatorname{tr}\big((D')'D'\big) = 0 \implies D' = 0 \implies \tilde\beta = \hat\beta & \quad \text{(Lecture 1, Theorem 10(c))}\\
\operatorname{var}(a'\tilde\beta) - \operatorname{var}(a'\hat\beta) = \sigma^2 a'DD'a = \sigma^2\|D'a\|^2 \geqslant 0 & \quad \text{(Lecture 6, Definition 1)}
\end{aligned}$$

$a = \operatorname{col}_j(I_p)$ gives $a'\tilde\beta = \tilde\beta_j$: $\operatorname{var}(\tilde\beta_j) \geqslant \operatorname{var}(\hat\beta_j)$.

□

Example 14. In Example 5, under (A1)–(A4): show that $c'y = \frac{1}{2}(y_1 - y_2)$ has $\mathsf{E}(c'y) = \beta_1$ for every β, and compare its variance with that of the entry $\hat\beta_1$ of the BLUE $\hat\beta$.

Solution.

$x_1' = (1,2,3)$, $x_2' = (1,0,3)$, $c = \frac{1}{2}(1,-1,0,0)'$, $a = (0,1,0)'$; (A1), (A3), Lecture 9, Theorem 3(b); Theorem 12(b) for $\operatorname{var}(\hat\beta_1)$; $M' = M$, $MX = 0$ (Theorem 10(c)):

$$\begin{aligned}
\mathsf{E}(y_1 - y_2) = (x_1 - x_2)'\beta = (0,2,0)\beta = 2\beta_1, \qquad \mathsf{E}(c'y) = \beta_1 = a'\beta & \\
\operatorname{var}(c'y) = \sigma^2 c'c = \sigma^2/2; \qquad \hat\beta_1 = \tfrac{1}{4}(y_1 - y_2 + y_3 - y_4), \quad \operatorname{var}(\hat\beta_1) = \sigma^2 \cdot \tfrac{1}{4} & \\
X'c = a, \qquad Hc = X(X'X)^{-1}X'c = X(X'X)^{-1}a = \tfrac{1}{4}(1,-1,1,-1)' & \\
u = c - Hc = Mc = \tfrac{1}{4}(1,-1,-1,1)' & \\
c' = a'(X'X)^{-1}X' + u', \qquad u'X = c'MX = 0 & \\
\operatorname{var}(c'y) - \operatorname{var}(\hat\beta_1) = \frac{\sigma^2}{2} - \frac{\sigma^2}{4} = \sigma^2\|u\|^2 = \sigma^2 \cdot \tfrac{4}{16} &
\end{aligned}$$

$\hat\beta_1$ is row 2 of $(X'X)^{-1}X'y$. For $y = (9,3,5,3)'$ the two estimates are $\frac{1}{2}(9-3) = 3$ and $\hat\beta_1 = 2$.

Proposition 15. Let $X \in \mathbb{R}^{n\times p}$, $y \in \mathbb{R}^n$, and $W \in \mathbb{R}^{n\times n}$ PD (Lecture 8, Definition 13: symmetric, $x'Wx > 0$ for $x \neq 0$). The minimizers of $Q(\beta) = (y - X\beta)'W(y - X\beta)$ ($\beta \in \mathbb{R}^p$) are the solutions of $X'WX\beta = X'Wy$. If $\operatorname{rank} X = p$, then $X'WX$ is PD, and the unique minimizer of Q is the **weighted least-squares estimate** $\hat{\beta}_W = (X'WX)^{-1}X'Wy$. For $W = \operatorname{diag}(w_1, \ldots, w_n)$ with all $w_i > 0$, $Q(\beta) = \sum_i w_i(y_i - x_i'\beta)^2$.

Proof.

$T = W^{1/2}$ (Lecture 10, Proposition 12: $W^{1/2}$, $W^{-1/2}$ symmetric, $W^{1/2}W^{1/2} = W$, $W^{1/2}W^{-1/2} = I$), so $T' = T$, $T'T = W$, and T is invertible (Lecture 2, Theorem 15):

$$
\begin{aligned}
Q(\beta) = (y - X\beta)'T'T(y - X\beta) = \|Ty - TX\beta\|^2 \quad & \text{(Lecture 1, Theorem 8; Lecture 6, Definition 1)} \\
\text{minimizers: } (TX)'TX\beta = (TX)'Ty \quad & \text{(Theorem 3(a) for } TX, Ty\text{)} \\
\iff X'WX\beta = X'Wy \quad & (T'T = W)
\end{aligned}
$$

$\operatorname{rank} X = p$:

$$
\begin{aligned}
\operatorname{rank}(TX) = \operatorname{rank} X = p \quad & \text{(Lecture 4, Theorem 15(d): } \operatorname{rank}(UAV) = \operatorname{rank} A,\ U, V \text{ invertible)} \\
X'WX = (TX)'(TX) \text{ is PD} \quad & \text{(Lecture 8, Theorem 15(b): } A'A \text{ is PD} \iff \operatorname{rank} A = p\text{)}
\end{aligned}
$$

The only minimizer is $\hat{\beta}_W = \left((TX)'TX\right)^{-1}(TX)'Ty = (X'WX)^{-1}X'Wy$ (Theorem 3(c) for TX, Ty).

For $W = \operatorname{diag}(w_1, \ldots, w_n)$, all $w_i > 0$: $W' = W$, $x'Wx = \sum_i w_i x_i^2 > 0$ for $x \neq 0$, so W is PD, and $Q(\beta) = \sum_i w_i u_i^2$ for $u = y - X\beta$ (Lecture 1, Definition 2, Proposition 13(a)). □

4. Estimating σ^2; sums of squares

Theorem 16. Assume (A2), (A3), and $\operatorname{rank} X = r < n$; let G be a g-inverse of $X'X$ and $M = I_n - XGX'$ (Theorem 10). Then $\mathsf{E}(\text{SSE}) = (n-r)\sigma^2$, so $\hat{\sigma}^2 = \text{SSE}/(n-r)$ is an unbiased estimator of σ^2. Under (A4), $r = p$, and $\hat{\sigma}^2$ is the **mean squared error** $s^2 = \text{MSE} = \text{SSE}/(n-p)$ ($= \text{SSE}/(n-k-1)$ in multiple regression).

Proof.

$y = X\beta + \varepsilon$; Lecture 9, Theorem 6: $\mathsf{E}(z'Az) = \operatorname{tr}(A\Sigma) + \mu'A\mu$ for $\mathsf{E}(z) = \mu$, $\operatorname{var}(z) = \Sigma$; here $z = \varepsilon$, $\mu = 0$ (A2), $\Sigma = \sigma^2 I_n$ (A3):

$$
\begin{aligned}
e = My = MX\beta + M\varepsilon = M\varepsilon \quad & \text{(Theorem 10(c),(e))} \\
\text{SSE} = e'e = \varepsilon'M'M\varepsilon = \varepsilon'M\varepsilon \quad & \text{(Theorem 10(c))} \\
\mathsf{E}(\text{SSE}) = \operatorname{tr}(M\sigma^2 I_n) + 0'M0 = \sigma^2 \operatorname{tr} M = \sigma^2(n-r) \quad & \text{(Lecture 1, Theorem 10(a); Theorem 10(d))} \\
\mathsf{E}(\hat{\sigma}^2) = \tfrac{1}{n-r}\,\mathsf{E}(\text{SSE}) = \sigma^2 \quad & (n - r > 0\text{; Lecture 9, Theorem 3(b))}
\end{aligned}
$$

□

Example 5: $\text{SSE} = S(\hat{\beta}) = 4$, $n - r = 4 - 3 = 1$, so $s^2 = 4$, and $s^2(X'X)^{-1} = \begin{pmatrix} 6 & -1 & -2 \\ -1 & 1 & 0 \\ -2 & 0 & 1 \end{pmatrix}$ estimates $\operatorname{var}(\hat{\beta})$ (Theorem 12(b)). Example 11: $\hat{\sigma}^2 = 4/(6-3) = \frac{4}{3}$.

Definition 17. For a least-squares fit, with $\hat{y}$ and $e = y - \hat{y}$: the **total**, **regression** and **error sums of squares** are $\text{SST} = \sum_i (y_i - \bar{y})^2$, $\text{SSR} = \sum_i (\hat{y}_i - \bar{y})^2$, $\text{SSE} = \sum_i (y_i - \hat{y}_i)^2 = e'e$. If $\mathbf{1}_n \in \mathcal{C}(X)$ and $\text{SST} > 0$, the **coefficient of determination** is $R^2 = \text{SSR}/\text{SST}$.

Theorem 18. If $\mathbf{1}_n \in \mathcal{C}(X)$: $\mathrm{SST} = \mathrm{SSR} + \mathrm{SSE}$, with $\mathrm{SST} = \|C_n y\|^2 = y' C_n y$ and $\mathrm{SSR} = \|C_n \hat{y}\|^2$ ($C_n = I_n - \frac{1}{n}\mathbf{1}\mathbf{1}'$, Lecture 1, Definition 14); if $\mathrm{SST} > 0$, $R^2 = 1 - \mathrm{SSE}/\mathrm{SST} \in [0,1]$; and $y'y = n\bar{y}^2 + \mathrm{SSR} + \mathrm{SSE}$.

Proof. $C_n' = C_n = C_n^2$, $C_n \mathbf{1} = 0$ (Lecture 1, Proposition 15); $C_n x = x - \bar{x}\mathbf{1}$ and $\sum_i (x_i - \bar{x})^2 = x' C_n x$ for $x \in \mathbb{R}^n$ (Lecture 1, Example 16).

$\hat{y}$ and y have the mean $\bar{y}$, $\mathbf{1}'e = 0$ (Proposition 4(b)):

$$
\begin{aligned}
\mathrm{SST} = \|C_n y\|^2 = y' C_n y, \qquad \mathrm{SSR} = \|C_n \hat{y}\|^2 \qquad & \text{(Lecture 1, Proposition 15, Example 16)} \\
C_n y = C_n \hat{y} + C_n e = C_n \hat{y} + e \qquad & (C_n e = e - \bar{e}\mathbf{1} = e) \\
(C_n \hat{y})' e = \hat{y}' C_n e = \hat{y}' e = 0 \qquad & \text{(Proposition 4(c))} \\
\mathrm{SST} = \|C_n \hat{y}\|^2 + \|e\|^2 = \mathrm{SSR} + \mathrm{SSE} \qquad & \text{(Lecture 6, Theorem 2: Pythagoras)} \\
\mathrm{SST} > 0 \implies R^2 = 1 - \mathrm{SSE}/\mathrm{SST}, \quad 0 \leqslant \mathrm{SSR} \leqslant \mathrm{SST} \qquad & (\mathrm{SSE} \geqslant 0) \\
y = \bar{y}\mathbf{1} + C_n y, \qquad (\bar{y}\mathbf{1})' C_n y = \bar{y}(C_n \mathbf{1})' y = 0 \qquad & \text{(Lecture 1, Proposition 15)} \\
y'y = n\bar{y}^2 + \|C_n y\|^2 = n\bar{y}^2 + \mathrm{SSR} + \mathrm{SSE} \qquad & \text{(Lecture 6, Theorem 2)}
\end{aligned}
$$

□

Example 19. In Example 5 ($\mathbf{1}_4$ is a column of X), find SST, SSR, SSE, R^2 and the partition of $y'y$.

Solution.

$\bar{y} = \frac{1}{4}(9+3+5+3) = 5$, $\hat{y} = (8,4,6,2)'$ (Example 5); Theorem 18:

$$
\begin{gathered}
C_4 y = (4,-2,0,-2)', \qquad C_4 \hat{y} = (3,-1,1,-3)', \qquad e = (1,-1,-1,1)', \qquad C_4 y = C_4 \hat{y} + e \\
\mathrm{SST} = 16+4+0+4 = 24, \quad \mathrm{SSR} = 9+1+1+9 = 20, \quad \mathrm{SSE} = 4: \quad 24 = 20 + 4, \quad R^2 = \tfrac{20}{24} = \tfrac{5}{6} \\
y'y = 81+9+25+9 = 124 = 4 \cdot 25 + 20 + 4 = 100 + 20 + 4
\end{gathered}
$$

Practice. The Exercises and Problems of the textbook, Chapters 12 and 13: generalized inverses, least squares and the normal equations, the hat matrix, rank-deficient designs, the Gauss–Markov theorem, the estimation of σ^2, and sums of squares.